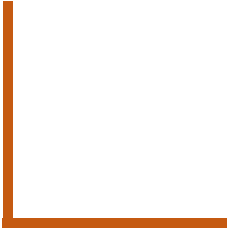




Design and Analysis of Algorithms

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DESIGN AND ANALYSIS OF ALGORITHMS

Important Problem Types

Slides courtesy of **Anany Levitin**

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Design and Analysis of Algorithms

Important Problem Types



- sorting
- searching
- string processing
- graph problems
- combinatorial problems
- geometric problems
- numerical problems

- Rearrange the items of a given list in ascending order.
 - Input: A sequence of n numbers $\langle a_1, a_2, \dots, a_n \rangle$
 - Output: A reordering $\langle a'_1, a'_2, \dots, a'_n \rangle$ of the input sequence such that $a'_1 \leq a'_2 \leq \dots \leq a'_n$.
- Why sorting?
 - Help searching
 - Algorithms often use sorting as a key subroutine.
- Sorting key

A specially chosen piece of information used to guide sorting.
Example: sort student records by SRN.

- Rearrange the items of a given list in ascending order.
- Examples of sorting algorithms
 - Selection sort
 - Bubble sort
 - Insertion sort
 - Merge sort
 - Heap sort ...
- Evaluate sorting algorithm complexity: the number of key comparisons.
- Two properties
 - Stability: A sorting algorithm is called stable if it preserves the relative order of any two equal elements in its input.
 - In place : A sorting algorithm is in place if it does not require extra memory, except, possibly for a few memory units.

Design and Analysis of Algorithms

Important Problem Types: Searching



Find a given value, called a **search key**, in a given set.

Examples of searching algorithms

- ▢ Sequential searching
- ▢ Binary searching...

A string is a sequence of characters from an alphabet.

Text strings: letters, numbers, and special characters.

String matching: searching for a given word/pattern in a text.

Text: I am a **computer** science graduate

Pattern: computer

Definition

Graph G is represented as a pair $G = (V, E)$,
where V is a finite set of vertices and E is a finite set of edges

Modeling real-life problems

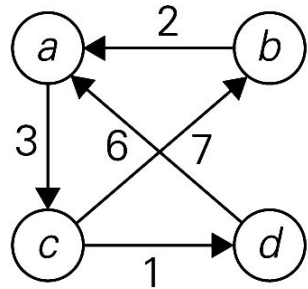
- Modeling WWW
- communication networks
- Project scheduling ...

Examples of graph algorithms

- Graph traversal algorithms
- Shortest-path algorithms
- Topological sorting

Shortest paths in a graph

To find the distances from each vertex to all other vertices.



(a)

$$W = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & \infty \\ \infty & 7 & 0 & 1 \\ 6 & \infty & \infty & 0 \end{bmatrix} \end{matrix}$$

(b)

$$D = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 10 & 3 & 4 \\ 2 & 0 & 5 & 6 \\ 7 & 7 & 0 & 1 \\ 6 & 16 & 9 & 0 \end{bmatrix} \end{matrix}$$

(c)

FIGURE 8.5 (a) Digraph. (b) Its weight matrix. (c) Its distance matrix.

Minimum cost spanning tree

- A spanning tree of a connected graph is its connected acyclic sub graph (i.e. a tree).

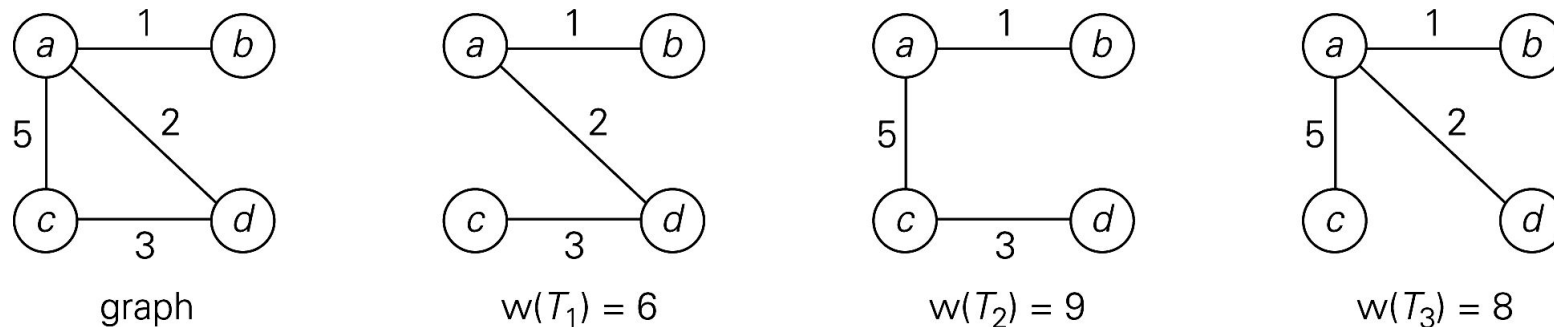
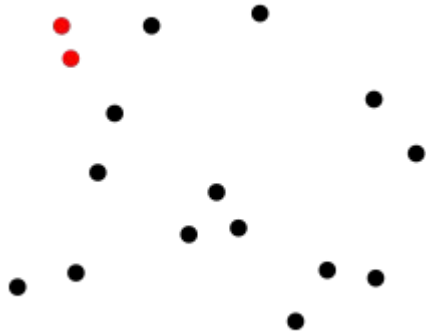
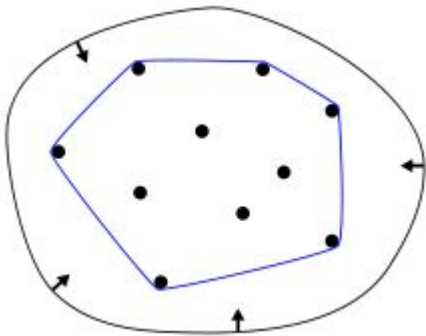


FIGURE 9.1 Graph and its spanning trees; T_1 is the minimum spanning tree

Closest Pair problem



Convex Hull Problem



- Solving Equations
- Computing definite integrals
- Evaluating functions

Which property of a sorting algorithm is essential when sorting student records by SRN while preserving the relative order of students with the same SRN?

- a) In-place
- b) Adaptive
- c) Stable
- d) Comparison-based

Answer

c) Stable

When analyzing the efficiency of comparison-based sorting algorithms, the primary performance metric emphasized in the slides is:

- a) Number of swaps
- b) Number of key comparisons
- c) Auxiliary space used
- d) Depth of recursion

Answer

- b) Number of key comparisons

Which statement best explains why binary search is asymptotically faster than sequential search?

- a) Binary search uses recursion
- b) Binary search performs fewer comparisons by halving the search space
- c) Binary search uses additional memory
- d) Binary search requires randomization

Answer

- c) Binary search performs fewer comparisons by halving the search space

Which real-world application best illustrates the use of graph algorithms?

- a) Sorting exam scores
- b) Finding the shortest route in a road network
- c) Searching a word in a document
- d) Computing convex hulls

Answer

- b) Finding the shortest route in a road network

The shortest-path problem in graphs primarily aims to:

- a) Find a path with minimum number of edges
- b) Determine shortest distances from one vertex to all other vertices
- c) Identify cycles in a graph
- c) Find the minimum spanning tree

Answer

- b) Determine shortest distances from one vertex to all other vertices

Why is a minimum spanning tree (MST) problem can be classified as a combinatorial problem?

- a) It involves numerical optimization
- b) It requires geometric computations
- c) It requires selecting an optimal subset of edges from many possible combinations
- d) It involves string manipulation

Answer

- b) It requires selecting an optimal subset of edges from many possible combinations

The convex hull problem is best described as finding:

- a) The shortest path between points
- b) The smallest enclosing circle of points
- c) The minimal convex polygon that encloses all given points
- d) The maximum spanning tree of points

Answer

- c) The minimal convex polygon that encloses all given points



THANK YOU

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